Hierarchical Pixel-Wavelength Fusion Network for Progressive Hyperspectral Image Reconstruction

Yuhao Yuan, Qun Zang, Zhiping Lin, Jincheng Liu, and Zhongmou Fan

***Abstract*— This study introduces the hierarchical pixel-wavelength fusion network (HPWF-Net), a novel hyperspectral reconstruction framework designed to address the challenges of tree species classification and spectral reconstruction in complex forest ecosystems. The proposed method employs a hierarchical pixel-wise and target-wavelength-wise reconstruction strategy, progressively recovering hyperspectral images by dynamically selecting and refining target wavelengths at each layer. HPWF-Net utilizes an enhanced U-Net architecture with multilevel loss supervision and convolutional consistency constraints, enabling accurate reconstruction by supervising intermediate encoder outputs, convolutional regressions, and final decoder predictions. This design preserves consistency with the original multispectral (MS) input while enhancing spectral fidelity and robustness. The model is trained on hyperspectral data from the NEON dataset and applied to Sentinel-2 MS imagery of the Daiyun Mountain National Nature Reserve. Experimental results demonstrate that the reconstructed hyperspectral data improve tree species classification accuracy by 5.9% compared to the original Sentinel-2 MS data. This work contributes to hyperspectral reconstruction and fine-grained ecological monitoring, offering a practical solution for remote sensing applications in forest ecosystems. Furthermore, we offer a Python-based package that enables to replicate our results at https://github.com/DarkFlameMaster02/Hierarchical-Pixel-Wavelength-Fusion-Network-for-Progressive-Hyperspectral-Image-Reconstruction**

***Index Terms*— Hierarchical reconstruction, hyperspectral image reconstruction, neural network for remote sensing.**

I. Introduction

FOREST ecosystems are integral to the global environment and the maintenance of ecological balance [1], [2]. The array of ecological services they provide extends beyond carbon sequestration, soil preservation, and water resource management to include critical functions such as biodiversity conservation and climate regulation [3], [4]. Tree species composition (TSC), as a fundamental metric of forest structure and function, not only impacts ecosystem productivity and stability but also offers essential insights for the evaluation and management of forest resources [5], [6], [7]. In the context of escalating climate change and increasing anthropogenic activities, a precise understanding and monitoring of TSC within forested areas have become imperative for effectively addressing ecological challenges [8], [9], [10].

Remote sensing technology has emerged as a pivotal tool for monitoring and analyzing forest ecosystems [11], [12], [13]. With extensive spatial coverage capabilities, remote sensing enables researchers to efficiently collect and analyze large-scale forest data, mitigating the temporal and logistical constraints associated with traditional ground surveys [14], [15], [16]. Recent advancements in remote sensing satellite technology, particularly the Landsat and Copernicus Sentinel series, have facilitated the utilization of multispectral (MS) and hyperspectral data [17], [18], capturing the spectral reflectance characteristics of tree species and significantly enhancing the accuracy of species identification [19]. However, despite its inherent advantages, remote sensing faces challenges in forest monitoring, particularly in addressing spatial resolution limitations and spectral similarities among tree species [20], [21], [22]. These limitations hinder precise mapping of TSC, especially in mixed and complex forest environments, where overlapping canopies and spectral heterogeneity prevail [23], [24]. Overcoming these challenges requires high-resolution spatial and spectral data, which enables a detailed understanding of species distribution patterns and ecosystem dynamics. Yet, due to technical and practical limitations, acquiring such high-resolution data remains challenging. This dilemma necessitates alternative strategies to enhance spectral discriminability from existing MS inputs, rather than solely pursuing higher resolution [25], [26].

To address the limitations of data availability, enhancing spectral resolution from existing MS imagery emerges as a viable pathway. While both spectral and spatial resolutions are critical, spectral refinement directly targets the biochemical properties of tree species, enabling discrimination of subtle interspecies differences even under mixed canopy conditions [27]. In contrast, purely increasing spatial resolution faces inherent constraints: structural complexity leads to persistent mixed pixels, and canopy occlusion limits understory spectral retrieval [28]. Thus, spectral enhancement offers a more robust foundation for accurate species mapping in heterogeneous forests. In contrast, spectral resolution enhancement directly addresses classification requirements by leveraging the biochemical properties of tree species, reflected

Received 23 December 2024; revised 4 March 2025 and 5 April 2025; accepted 30 April 2025. Date of publication 6 May 2025; date of current version 15 May 2025. This work was supported in part by the Ministry of Science and Technology of China under the National Natural Science Foundation of China under Grant 32101523 and Grant 32371875 and in part by Fujian Provincial Natural Science Foundation of China under Grant 2023J01080. *(Corresponding author: Zhongmou Fan.)*

Yuhao Yuan, Qun Zang, Zhiping Lin, and Zhongmou Fan are with the College of Transportation and Civil Engineering, Fujian Agriculture and Forestry University, Fuzhou 350002, China (e-mail: fujtfan2016@fafu.edu.cn).

Jincheng Liu is with the College of Natural Resources and Environment, Northwest A&F University, Yangling 712100, China.

Digital Object Identifier 10.1109/TGRS.2025.3567469

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

in their absorption and reflectance characteristics at specific wavelengths [29], [30]. By reconstructing hyperspectral data from MS inputs, subtle interspecies differences in biochemical traits (e.g., chlorophyll and lignin) can be captured, even among taxonomically close species [31], [32]. This capability mitigates spectral similarity challenges and provides a physics-grounded approach to species discrimination.

Current approaches to hyperspectral reconstruction in remote sensing are broadly categorized into endmember-based methods and neural network-based mapping methods. Endmember-based methods rely on spectral unmixing techniques, treating observed mixed pixel spectra as a combination of representative endmember spectra [33], [34]. These methods calculate endmember abundances to reconstruct the target spectra, providing strong physical interpretability [35]. However, they face significant challenges in complex ecosystems, such as mixed forests. The intricate spatial distribution of species complicates the accurate confirmation of species composition, often requiring ground-based surveys or auxiliary data, which introduces logistical challenges and increases uncertainties [36]. Furthermore, standard endmember libraries are often inadequate for capturing the spectral variability of real-world conditions, such as differences within the same species across growth stages or environmental contexts [37]. These limitations reduce the reconstruction accuracy and scalability of endmember-based approaches, particularly in dynamic environments where frequent updates to endmember libraries are necessary.

Neural network-based methods, in contrast, utilize data-driven models to predict hyperspectral bands from limited inputs, such as RGB channels [38], [39], [40]. These methods eliminate the need for endmember dictionaries and leverage the high-dimensional mapping capabilities of neural networks [41]. However, they often fail to explicitly model the intrinsic relationships between known and target bands, which in remote sensing are governed by physical and chemical principles. Additionally, these methods require large-scale datasets for effective training, and significant spectral variability across different scenes limits their generalization capability [31]. These challenges highlight the need for innovative approaches that integrate domain-specific spectral relationships, reduce dependency on extensive training data, and maintain robust performance across diverse ecosystems.

Given the unique challenges of mapping mixed forests and the limitations of existing methods, this study introduces an innovative approach designed to address the complexities of hyperspectral reconstruction and classification. The proposed framework incorporates the following key contributions.

1) *Pixel- Wavelength Integration for Enhanced Spectral Reconstruction:* Recognizing that each target wavelength possesses unique spectral characteristics, the model treats per-pixel features as shared foundational representations while refining these features for each target wavelength. This approach ensures that both shared and wavelength-specific information are effectively utilized, enabling more accurate hyperspectral reconstruction.
2) *Dynamic Spectral Selection in Progressive Reconstruction:* By extending the U-Net backbone, we propose an adaptive wavelength selection mechanism that dynamically optimizes the selection of target spectral bands during each reconstruction stage. Through spectral proximity analysis, each network layer autonomously identifies the most reconstructible subset of wavelengths from candidate bands, guided by evolving spectral context. This dynamic selection-generation framework achieves dual optimization, including progressive spectral refinement through prioritized reconstruction of spectrally adjacent bands to establish a robust physical foundation and adaptive error correction via multilevel supervision.
3) *Application to Mixed Forest Mapping:* The framework is applied to generate hyperspectral images for a study area, enabling pixel-wise classification and providing detailed insights for fine-scale forest mapping. This application demonstrates the practical utility of the proposed method in addressing the challenges of species-level forest composition analysis in complex ecosystems.

This approach not only addresses the spectral variability and high correlation inherent to hyperspectral data but also enhances the robustness and interpretability of the reconstructed outputs, paving the way for more accurate and detailed forest mapping under challenging conditions.

II. Study Area and Material

A. Study Area

The study focuses on the Daiyun Mountain National Nature Reserve, located in central Fujian, within Dehua, Quanzhou. This area encompasses the central section of the Daiyun Mountain range, including its main peak and surrounding regions. Geographically, the reserve is positioned between 118°05′22″E and 118°20′15″E longitude, and 25°38′07″N–25°43′40″N latitude (see Fig. 1). The terrain is complex, and the ecological environment is diverse. The reserve is representative of a tropical monsoon evergreen broadleaf forest ecosystem, characterized by significant regional ecological diversity and complex biological communities [42]. The Daiyun Mountain Nature Reserve covers 13 472.4 hectares and is divided into three functional zones: the core zone (5514.1 hectares), buffer zone (3515.4 hectares), and experimental zone (4442.9 hectares) [43]. The region has a temperate and humid climate, with consistent and abundant annual precipitation, which contributes to the high stability and biodiversity of the forest ecosystem. Forest coverage is notably high, reaching 93.4% [see Fig. 1(a)]. The vegetation in the reserve is categorized into nine distinct types, reflecting its ecological complexity and making the area an invaluable natural laboratory for biodiversity conservation and ecosystem research.

According to the 2018 forest resource survey, the reserve is subdivided into 1958 forest stands, encompassing a wide variety of land types, including pure forests, mixed forests, shrublands, and residential areas. These forest stands vary greatly in size, with the largest plot covering 801 464 m^2 and the smallest just 92 m^2 [see Fig. 1(b)]. The detailed spatial distribution of these plots provides a robust dataset for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

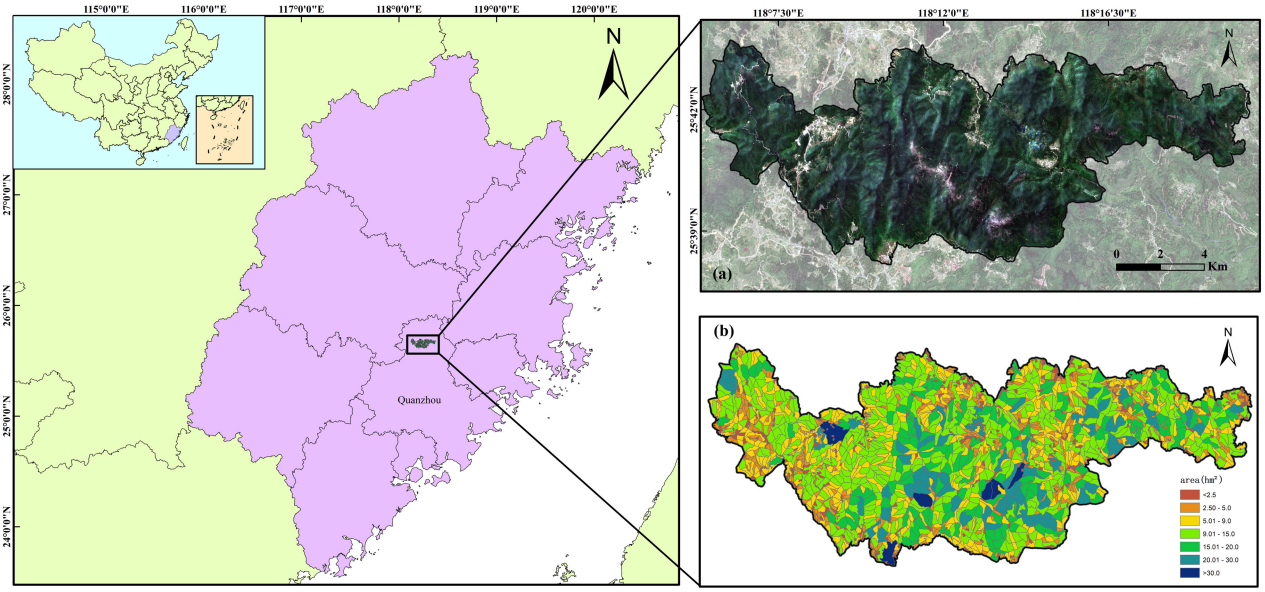


Fig. 1. Geographic location (a) and zonal divisions (b) of the Daiyun Mountain National Nature Reserve.

analyzing ecosystem characteristics and TSC in subsequent studies.

B. Data Sources and Characteristics

This study utilizes a comprehensive dataset, including hyperspectral data from the NEON dataset for training a reconstruction model, and MS data and forest detailed mapping data from the study area to evaluate the classification performance of the original MS data and the spectrally reconstructed high spectral data, which uses pretrained models.

1) Hyperspectral Data From NEON: The hyperspectral data (see Fig. 2) were obtained from the NEON dataset [44], [45], which provided high-dimensional spectral information essential for targeted training of the proposed hyperspectral reconstruction model for mixed forests, enabling the capture of subtle spectral variations within complex forest ecosystems.

Specifically, the data were acquired by NEON's Airborne Observation Platform (AOP) between June 28 and July 6, 2017, in the D17 Pacific Southwest study region. Hyperspectral data were captured using the AVIRIS next-generation push-broom instrument, developed by NASA's Jet Propulsion Laboratory, at a flight altitude of approximately 1000 m. This instrument measures reflectance across 426 spectral bands with a wavelength range of 380–2510 nm and a spectral sampling interval of 5 nm. The imagery was collected with a cross-track field of view of 34° and an instantaneous field of view of 1.0 mrad, resulting in a ground resolution of approximately 1 m. The raw data underwent atmospheric correction using ATCOR4r and orthorectification with nearest neighbor resampling, aligning the images to a fixed and uniform spatial grid.

2) MS Data and Forest Mapping Data From Study Area: MS imagery for this study was acquired from the Copernicus Open Access Hub [46], specifically utilizing Sentinel-2 Level 2A data products (see Fig. 1), and the imaging time was April 9, 2018. Forest mapping data were derived from the

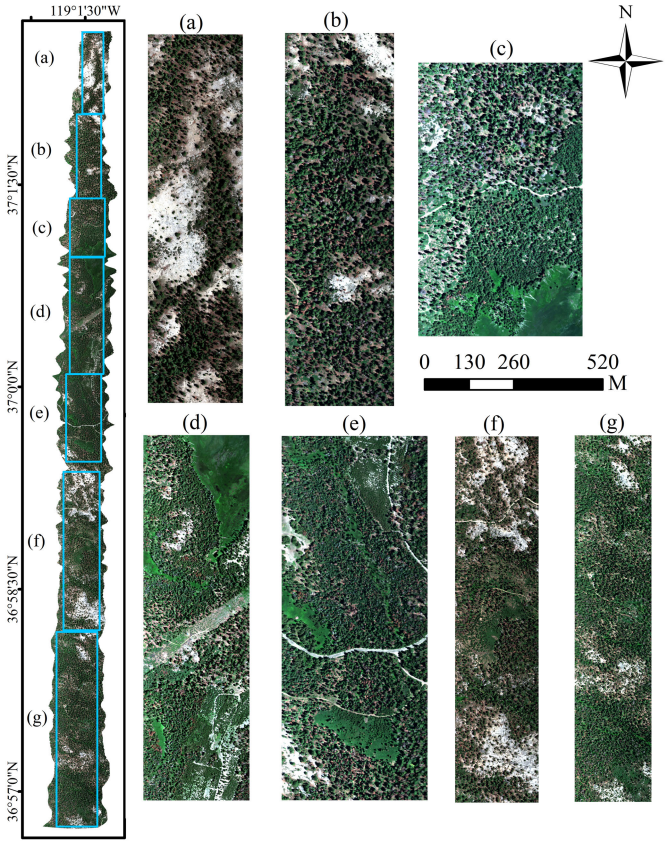


Fig. 2. Hyperspectral imagery (a)–(g) provided by the NEON dataset. (a)–(g) Used for hyperspectral image reconstruction train.

National Forest Resource Second-Class Inventory [47]. The integration of these diverse datasets facilitated a comprehensive assessment of the model's reconstruction capabilities and demonstrates its potential for advancing spectral analysis in mixed and complex forest ecosystems.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

C. Data Preprocessing

To ensure the quality and consistency of the data, the data preprocessing in this study is divided into two main parts: hyperspectral data preprocessing and study area data preprocessing. The preprocessing of hyperspectral data is aimed at evaluating the performance of the proposed reconstruction model, while the preprocessing of study area data is intended to validate the generalization ability of the model. By applying the pretrained model to the MS data in the study area for spectral enhancement and comparing the classification performance of the hyperspectral and MS datasets, the significance of spectral reconstruction in remote sensing imagery is demonstrated. These preprocessing steps are crucial for aligning the data and ensuring that the model accurately reflects the ecological diversity of the study area, thereby enhancing model training and its classification application evaluation in real-world scenarios.

1) Hyperspectral Data Preprocessing: In this study, the preprocessing of hyperspectral data begins by applying the spectral response function (SRF) of Sentinel-2 to convert the original hyperspectral data. The compression process of SRF on hyperspectral data can be expressed in two ways: one is in the form of an integral and the other is through discrete matrix multiplication.

First, the continuous integral form of SRF describes how the reflectance for each target band is obtained by weighted integration. Let $X_{\text{hyperspectral}}$ represent the hyperspectral data, which contains multiple continuous bands $X_{\text{hyperspectral},i}$, and the integral form of SRF can be expressed as

$$R_{\text{Sentinel},j} = \int_{-\infty}^{\infty} X_{\text{hyperspectral},i} \cdot S_{ij} d\lambda \tag{1}$$

where $R_{\text{Sentinel},j}$ is the reflectance of the jth Sentinel-2 band; $X_{\text{hyperspectral},i}$ is the reflectance of the ith band in the hyperspectral data; S_{ij} is the response coefficient in the SRF matrix, representing the relationship between the ith hyperspectral band and the jth Sentinel-2 band; and λ is the wavelength variable.

However, in actual remote sensing data processing, hyperspectral data are usually discrete. Therefore, the integral form of SRF is discretized into matrix multiplication. Let $X_{\text{hyperspectral}} \in \mathbb{R}^{N\times M}$, where N is the number of pixels and M is the number of hyperspectral bands. The SRF matrix $S \in \mathbb{R}^{M\times K}$ represents the number of hyperspectral bands M and the target bands K (which in this study is 13, corresponding to Sentinel-2 bands). The discretized compression process is realized by the following matrix multiplication:

$$X_{\text{Sentinel}-2} = X_{\text{hyperspectral}} \times S \tag{2}$$

where $X_{\text{Sentinel}-2} \in \mathbb{R}^{N\times K}$ is the resulting data matrix, representing the reflectance of each pixel in the 13 Sentinel-2 bands.

In this discrete matrix multiplication, the SRF acts to map the reflectance of hyperspectral bands to the reflectance of target bands. Specifically, each row S_i in the matrix corresponds to the response weights between the ith hyperspectral band and the target bands. Through this matrix multiplication, the high-dimensional information from the original hyperspectral data is effectively compressed into a low-dimensional space, generating data similar to Sentinel-2.

Based on this, we select the four bands (B2, B3, B4, and B8) with a 10-m spatial resolution from the simulated Sentinel-2 data as the sample data. The original hyperspectral data from 490 to 840 nm (with a 5-nm interval, totaling 71 bands) are used as the label data. This forms a sample–label pair, including $X_{\text{samples}} \in \mathbb{R}^{N\times 4}$ and $X_{\text{labels}} \in \mathbb{R}^{N\times 71}$, where X_{samples} represents the Sentinel-2 sample data after SRF transformation, consisting of the four 10-m resolution bands, and X_{labels} represents the 71 bands selected from the original hyperspectral data as the label data.

Through this preprocessing step, we effectively convert the hyperspectral data into a format similar to Sentinel-2, generating sample–label pairs for subsequent model training. This process not only ensures data consistency but also provides the necessary input for efficient model training and evaluation.

2) Study Area Data Preprocessing: The preprocessing of study area data mainly involves the processing of MS data and forest mapping data. The MS data for the study area were obtained from Sentinel-2 through the Copernicus Open Access Hub. To ensure the validity and accuracy of the data, all images underwent a strict visual screening process to remove cloud contamination. The data were then cropped to align precisely with the study area boundaries, ensuring spatial consistency and data alignment.

The forest mapping data for the study area were sourced from the National Forest Resource Second-Class Inventory, specifically using the shapefile of pure forests. By applying these shapefiles to the remote sensing images of the study area, we were able to crop the images accurately, establishing a correspondence between pixels and different forest types. This process provides the necessary label information for model validation, ensuring that the training data align with the actual characteristics of the area.

Through these preprocessing steps, the MS data and forest mapping data for the study area were effectively aligned, providing a solid data foundation for the subsequent evaluation of the model's generalization ability and classification performance.

III. Method

In this section, we provide a detailed description of the specific methodologies employed in this study, focusing solely on the framework and implementation of the model. The key steps and structure of each method are outlined, along with their application within the context of this research.

A. Problem Formulation

Hyperspectral images record the reflectance information of the ground over continuous wavelengths, while MS images capture the average radiative energy within broader spectral ranges [48]. Their relationship can be expressed as follows; for hyperspectral data, the energy $H_i(p)$ at a specific wavelength λ_i is given by

$$H_i(p) = \int L(p,\lambda) S_i(\lambda) d\lambda \approx L(p,\lambda_i) \tag{3}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

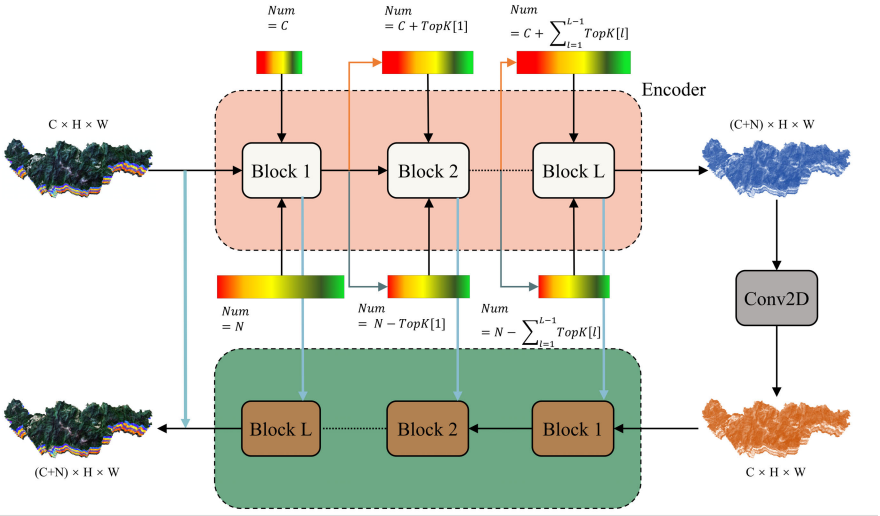


Fig. 3. Overall architecture of the proposed HPWF-Net, based on an improved U-Net structure.

where $S_i(\lambda)$ is the SRF of the hyperspectral sensor, typically approximated as a narrow pulse centered at λ_i. For MS data, the energy $M_k(p)$ recorded by a wideband sensor is represented as

$$M_k(p) = \int L(p,\eta) S_k(\eta) d\eta \tag{4}$$

where $S_k(\eta)$ denotes the SRF of the MS sensor, covering a broader wavelength range. Based on this relationship, a parameterized function can be designed to establish a mapping between MS and hyperspectral data

$$H_i(p) = f\big(M(p), \lambda_i; \theta\big) \tag{5}$$

where f is a nonlinear function with neural network model parameters θ that extracts its feature vectors relative to known wavelengths from MS data $M(p)$ based on a given wavelength λ_i and predicts the hyperspectral data $H_i(p)$.

B. Network Architecture

Fig. 3 illustrates the overall architecture of the proposed hierarchical pixel-wavelength fusion network (HPWF-Net), which is based on a U-Net architecture [51], [52], designed for spectral reconstruction tasks. The network comprises three main components: the encoder, intermediate convolutional constraint (Conv2D) [53], and the decoder. These components collectively perform hierarchical pixel-wavelength feature fusion while preserving spatial resolution and ensuring reconstruction consistency [54].

The encoder progressively reconstructs hierarchical features from the input MS image $I_{\text{MSI}} \in \mathbb{R}^{C\times H\times W}$, where C is the number of MS bands and $H \times W$ represents the spatial dimensions. The number of blocks in the encoder is dynamic and can be adjusted based on the specific task. At each block, the wavelengths to be reconstructed are progressively determined, with the previously reconstructed wavelengths combined with the original known wavelengths from the input as pseudo-known wavelengths for the reconstruction of the next set of target wavelengths. In the first block, the model reconstructs the wavelengths most relevant to the known wavelengths, and for each subsequent block, the previously reconstructed wavelengths are incorporated as additional pseudo-known wavelengths for the reconstruction of the next set of target wavelengths. The number of specific wavelengths to be reconstructed at each block is not fixed but is represented by the array Top$K$, where Top$K[i]$ indicates the wavelengths to be reconstructed at the ith block. Formally, the output of the ith encoder block can be expressed as

$$F^{(i)} = \text{Block}\big(F^{(i-1)}\big) \tag{6}$$

where $F^{(i)} \in \mathbb{R}^{(C+\sum_{l=1}^{i}\text{Top}K[l])\times H\times W}$, for $i \geq 1$ and $F^{(0)} = I_{\text{MSI}}$. The final output of the encoder is $F^{(L)}$, where L represents the total number of block and has $C + N$ spectral dimensions, with N being the total number of target wavelengths.

To address potential information loss in U-Net architectures [55], the intermediate encoder output undergoes a convolutional operation, restoring its spectral dimensions to C. This step acts as a constraint to enforce alignment with the original MS input, preserving essential details throughout the reconstruction process as

$$F_{\text{conv}} = g\big(F^{(L)}\big), \quad F_{\text{conv}} \in \mathbb{R}^{C\times H\times W} \tag{7}$$

where $g(\cdot)$ represents the Conv2D operation applied to the encoder's final output.

The decoder mirrors the encoder's structure and reconstructs the feature maps into the final hyperspectral image as $I_{\text{HSI}} \in \mathbb{R}^{(C+N)\times H\times W}$. To preserve fine-grained spatial and spectral information, skip connections are introduced between the encoder and the decoder. These skip connections [56] utilize the output of specific components within each encoder block, capturing detailed spatial–spectral features and passing them to the corresponding stages in the decoder

$$I_{\text{HSI}} = \text{Decoder}(F_{\text{conv}}, \text{Skip connections}). \tag{8}$$

The HPWF-Net reconstructs hyperspectral data by combining dynamic reconstruction of hierarchical features, intermediate constraints, and skip connections. The encoder

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

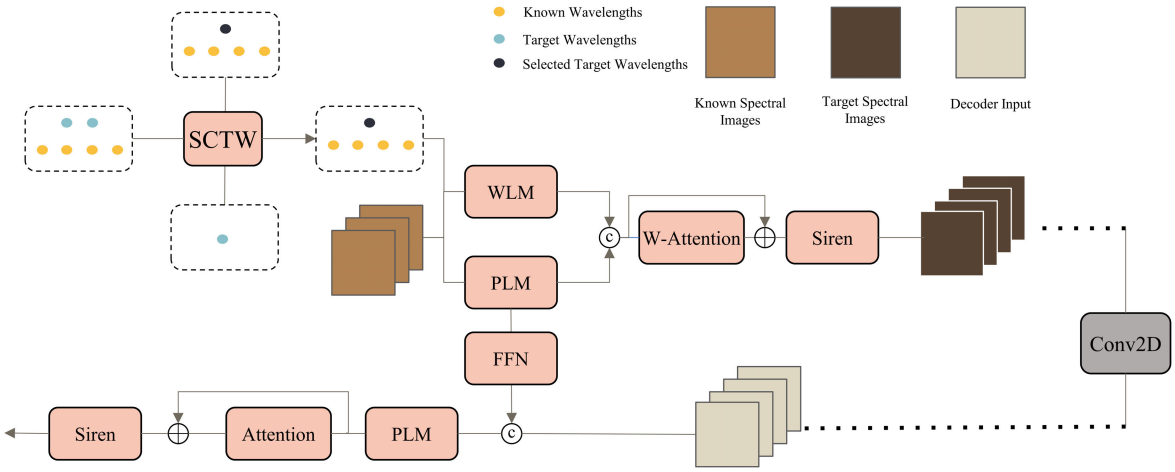


Fig. 4. Detailed structure of the encoder and decoder blocks, illustrating the feature extraction process of the WLM and PLM.

progressively enhances pixel and wavelength-level features, while the intermediate Conv2D operation ensures consistency with the original MS input. The decoder mirrors the encoder's structure, integrating these enhanced features through skip connections that preserve fine-grained spatial and spectral information, ultimately producing high-quality hyperspectral reconstructions.

C. Detail of Encoder Block and Decoder Block

The encoder block and decoder block are the core components of the HPWF-Net, designed to hierarchically extract features across both pixel and wavelength dimensions, progressively enabling hyperspectral information reconstruction (see Algorithms 1 and 2). Fig. 4 illustrates the overall structure of these modules and the flow of information within them.

Algorithm 1 Detail of the ith Encoder Block

Input:
Pseudo Known Wavelength Image $\boldsymbol{F^{(i-1)}} \in \mathbb{R}^{(C+\sum_{l=1}^{i-1} TopK[l])\times H\times W}$,
Candidate Target Wavelengths $\boldsymbol{T_{candidate}} \in \mathbb{N}^{N-(C+\sum_{l=1}^{i-1} TopK[l])}$,
Current Known Wavelengths $\boldsymbol{K_{known}} \in \mathbb{N}^{(C+\sum_{l=1}^{i-1} TopK[l])}$
Number of wavelengths to be reconstructed in the i-th block $\boldsymbol{TopK[i]}$
Output: Block output $\boldsymbol{F^{(i)}} \in \mathbb{R}^{(C+\sum_{l=1}^{i} TopK[l])\times H\times W}$,
Skip Connection $\boldsymbol{Skip} \in \mathbb{R}^{TopK[i]\times H\times W}$,
Candidate Target Wavelengths of Next Block
$\boldsymbol{T_{candidate,next}} \in \mathbb{N}^{N-(C+\sum_{l=1}^{i} TopK[l])}$,
Known Wavelengths of Next Block $\boldsymbol{K_{next}} \in \mathbb{N}^{(C+\sum_{l=1}^{i} TopK[l])}$
$\#T_{selected} \in \mathbb{N}^{\boldsymbol{TopK[i]}}$
1. $\boldsymbol{T_{candidate,next}}, \boldsymbol{K_{next}}, T_{selected} = SCTW(\boldsymbol{T_{candidate}}, \boldsymbol{K_{known}}, \boldsymbol{TopK[i]})$
Extract pixel-level features without spectral expansion
2. $\boldsymbol{Skip} = PLM(\boldsymbol{F^{(i-1)}}).No_Expanded.FFN$
$\#Feature\ and\ Feature_{Enhanced} \in \mathbb{R}^{(C+4+\sum_{l=1}^{i-1} TopK[l])\times H\times W\times TopK[i]}$
3. $Feature = Concat(WLM(T_{selected}, \boldsymbol{F^{(i-1)}}, \boldsymbol{K_{known}}), PLM(\boldsymbol{F^{(i-1)}}))$
4. $Feature_{Enhanced} = Feature + WavelengthAttention(Feature)$
5. $\boldsymbol{F^{(i)}} = Siren(Feature_{Enhanced})$
6. **return** $\boldsymbol{F^{(i)}}$, $\boldsymbol{Skip}$, $\boldsymbol{T_{candidate-NextBlock}}$, $\boldsymbol{K_{known-NextBlock}}$

Algorithm 2 Detail of the ith Decoder Block

Input: From previous decoder block (i−1) or middle bottleneck $\boldsymbol{X} \in \mathbb{R}^{(C+\sum_{l=1}^{i-1} TopK[L-l+1])\times H\times W}$,
Skip Connection $\boldsymbol{Skip} \in \mathbb{R}^{TopK[L-i+1]\times H\times W}$
Output: the i-th Decoder Block output $\boldsymbol{output} \in \mathbb{R}^{(C+\sum_{l=1}^{i} TopK[L-l+1])\times H\times W}$
1. $Feature = Attention(PLM_{Decoder}(Integrate(\boldsymbol{X}, \boldsymbol{Skip})))$
2. $\boldsymbol{output} = Siren(Feature)$
3. **return** $\boldsymbol{output}$

In the encoder block, spectral and spatial features are extracted through a carefully designed sequence of operations. We first employ the select current target wavelength (SCTW) module to identify the target wavelengths to be reconstructed at the current layer, serving as the foundation for subsequent feature extraction. The wavelength-level module (WLM) captures the spectral relationships between target wavelengths and their neighboring known wavelengths, providing localized spectral context for target-wavelength representation. The pixel-level module (PLM) focuses on extracting spatial features from the local pixel neighborhood. These spatial features are expanded and concatenated with the outputs of the WLM to form a unified representation that integrates both spectral information and spatial information. This combined representation is further processed by the wavelength attention module, which captures dependencies among target wavelengths, followed by the Siren network to predict the spectral values of the target wavelengths.

The decoder block integrates the hierarchical features from the previous layer with skip connections from the encoder, enabling further refinement of global and local features. The combined features are processed through the PLM and a standard self-attention mechanism, enhancing the consistency and expressiveness of both global and local information. Finally, these features are passed through the Siren network

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

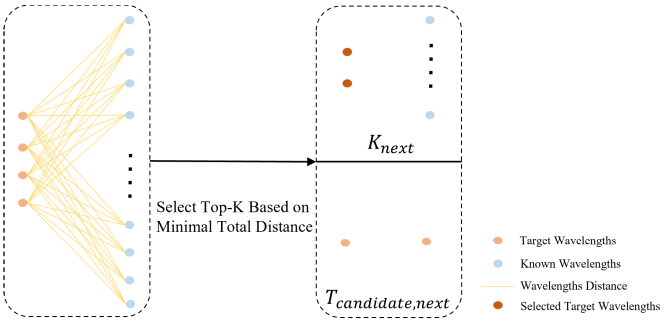


Fig. 5. Flow diagram of the SCTW module, demonstrating the dynamic selection of target wavelengths within the spectral domain.

to reconstruct the spectral values for the target wavelengths in a layer-by-layer manner.

This design leverages the hierarchical feature extraction of the encoder block and the integrative capability of the decoder block to achieve deep modeling of spectral and spatial features, significantly improving the accuracy and robustness of hyperspectral image reconstruction.

D. Module Details

1) SCTW Module: The SCTW module is designed to dynamically determine the target wavelengths for reconstruction at each encoding layer, enabling a progressive and structured approach to spectral reconstruction, similar in concept to hierarchical frameworks like HRNet [57]. However, unlike HRNet's spatial hierarchy, SCTW enforces spectral hierarchy by dynamically updating known wavelengths, ensuring progressive refinement of spectral context. SCTW employs a distinct wavelength selection mechanism tailored for spectral tasks (see Algorithm 3). As illustrated in Fig. 5, SCTW evaluates the relationship between candidate target wavelengths and the known wavelengths. Specifically, for each candidate target wavelength, SCTW computes the total distance to all known wavelengths in the spectral domain, where the distance $d(t, k)$ encodes the spectral proximity, represents the wavelength difference, and quantifies how closely the target wavelength t is related to the known wavelength k.

Algorithm 3 Select Current Target Wavelength Module for the ith Encoder Block

Input: Candidate Target Wavelengths $\boldsymbol{T_{candidate}}$,
Current Known Wavelengths $\boldsymbol{K_{known}}$,
Number of Wavelengths that need to be rebuilt $\boldsymbol{TopK[i]}$
Output: Selected Wavelengths $\boldsymbol{T_{selected}} \in \mathbb{N}^{\boldsymbol{TopK[i]}}$, Updated known set $\boldsymbol{K_{next}}$, Updated candidate set $\boldsymbol{T_{candidate,next}}$
$\mathbb{S} = \{T \subseteq \boldsymbol{T_{candidate}} || T| = \boldsymbol{TopK[i]}\}$
for each T in $\mathbb{S}$ do
  $D(T) = \sum_{t\in T} \sum_{k\in \boldsymbol{K_{known}}} d(t, k)$
end for
$\boldsymbol{T_{selected}} = \underset{T\subseteq\mathbb{S}}{argmin}\, D(T)$
$\boldsymbol{K_{next}} = \boldsymbol{K_{known}} \cup \boldsymbol{T_{selected}}$
$\boldsymbol{T_{candidate,next}} = \boldsymbol{T_{candidate}} \setminus \boldsymbol{T_{selected}}$
return $\boldsymbol{T_{selected}}$, $\boldsymbol{K_{next}}$, $\boldsymbol{T_{candidate,next}}$

Based on the computed total distances, SCTW selects Top$K[i]$ target wavelengths with the smallest cumulative distances, expressed mathematically as

$$T_{\text{selected}} = \underset{T\subseteq T_{\text{candidate}}, |T|=\text{Top}K[i]}{\text{argmin}} \sum_{t\in T} \sum_{k\in K_{\text{known}}} d(t, k) \tag{9}$$

where $T_{\text{candidate}}$ represents the set of candidate target wavelengths and K_{known} represents the set of known wavelengths.

The selected wavelengths (T_{selected}) are treated as the current layer's target wavelengths for prediction and, upon completion of this layer, are merged with the existing known wavelengths to form the set of known wavelengths for the next layer

$$K_{\text{next}} = K_{\text{known}} \cup T_{\text{selected}}. \tag{10}$$

The remaining candidate wavelengths, excluding the selected ones, form the new set of candidate target wavelengths for the next layer

$$T_{\text{candidate,next}} = T_{\text{candidate}} \backslash T_{\text{selected}}. \tag{11}$$

This progressive strategy ensures that the encoder focuses on reconstructing the most contextually relevant wavelengths at each layer while simultaneously updating the spectral context for subsequent layers. By iteratively refining the target-wavelength selection, SCTW effectively supports the hierarchical reconstruction of hyperspectral data, balancing local spectral relationships and global reconstruction objectives.

Algorithm 4 Wavelength-Level Module for the ith Block

Input: Current Target Wavelength $\lambda_t \subseteq \boldsymbol{T_{selected}}$,
spectral reflectance values $\boldsymbol{r}(\lambda_{k1}), \boldsymbol{r}(\lambda_{k2}) \subseteq \boldsymbol{F^{(i-1)}}$,
Current Known Wavelengths $\boldsymbol{K_{known}}$
Output: Feature vector $\boldsymbol{f}(\lambda_t) \in \mathbb{R}^4$
$\lambda_{k1}, \lambda_{k2} = Neighbor(\lambda_t, \boldsymbol{K_{known}})$
$p(\lambda_t) = \frac{\lambda_t - \lambda_{k1}}{\lambda_{k2} - \lambda_{k1}}$
$f(\lambda_t) = [\lambda_t, \boldsymbol{r}(\lambda_{k1}), \boldsymbol{r}(\lambda_{k2}), p(\lambda_t)]$
return $\boldsymbol{f}(\lambda_t)$

2) WLM: The WLM (see Fig. 6) is designed to provide each target wavelength with a comprehensive representation by capturing its spectral context relative to the known wavelengths (see Algorithm 4). For each target wavelength λ_t, the module identifies the interval $[\lambda_{k1}, \lambda_{k2}]$, where λ_{k1} and λ_{k2} are the closest known wavelengths that bound λ_t in the spectral domain. The module constructs a feature vector $f(\lambda_t)$ that integrates spectral and positional information, expressed as

$$f(\lambda_t) = [\lambda_t, r(\lambda_{k1}), r(\lambda_{k2}), p(\lambda_t)] \tag{12}$$

where λ_t is the target wavelength, $r(\lambda_{k1})$ and $r(\lambda_{k2})$ are the spectral reflectance values at the bounding wavelengths, and $p(\lambda_t)$ represents the relative position of the target wavelength within the interval. The relative position is computed as

$$p(\lambda_t) = \frac{\lambda_t - \lambda_{k1}}{\lambda_{k2} - \lambda_{k1}}. \tag{13}$$

This formulation ensures that each target wavelength is equipped with a contextually enriched feature representation, which encapsulates its positional relationship and spectral proximity to the neighboring known wavelengths. The structured feature vector $f(\lambda_t)$ is used as input for subsequent

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

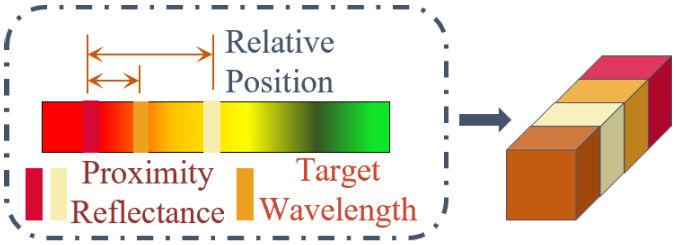


Fig. 6. Structure of the WLM, capturing spectral relationships between target wavelengths and neighboring known wavelengths.

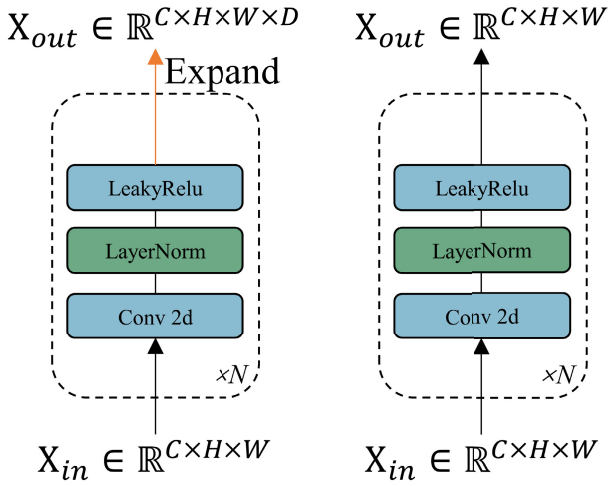


Fig. 7. Structure of the PLM, showing the spatial feature extraction and expansion process in the encoder and decoder.

stages, enabling the model to effectively leverage local spectral relationships to enhance the reconstruction accuracy and robustness.

3) PLM: The PLM (see Fig. 7) performs pixel-level feature extraction through a series of convolutional operations, with its behavior adapting based on its placement within the encoder or decoder. In the encoder, the PLM refines pixel-level features and subsequently expands them along the wavelength dimension, aligning the spatial features with the wavelength-specific features from the WLM. Starting from an input $X_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, the PLM processes the features through N convolutional layers and expands the output to $X_{\text{out}} \in \mathbb{R}^{C\times H\times W\times D}$, where D is the number of target wavelengths. This expanded representation ensures seamless fusion of spatial and spectral features, facilitating effective reconstruction.

In the decoder, however, the PLM focuses on refining pixel-level features for reconstruction tasks and directly outputs $X_{\text{out}} \in \mathbb{R}^{C\times H\times W}$ without further expansion. This design maintains computational efficiency while preserving essential spatial details for decoding, aligning the spatial context directly with the target wavelengths reconstructed in the decoder. By adapting its functionality within different stages, the PLM ensures effective integration of spatial and spectral information throughout the reconstruction process.

Algorithm 5 Wavelength Attention Module

Input: Original wavelength feature $\boldsymbol{X_{in}} \in \mathbb{R}^{D\times H\times W\times(C+4)}$
Output: Refined wavelength feature $\boldsymbol{X_{out}} \in \mathbb{R}^{D\times H\times W\times(C+4)}$
1: $X = Reshape(\boldsymbol{X_{in}}) \# X \in \mathbb{R}^{(H\times W)\times D\times(C+4)}$
2: $Q = X \cdot W^Q, K = X \cdot W^K, V = X \cdot W^V$
3: $Attention = Softmax(\frac{QK^T}{\sqrt{d_k}})$
4: $\boldsymbol{X_{out}} = Reshape(Attention \cdot V)$
5: **return** $\boldsymbol{X_{out}}$

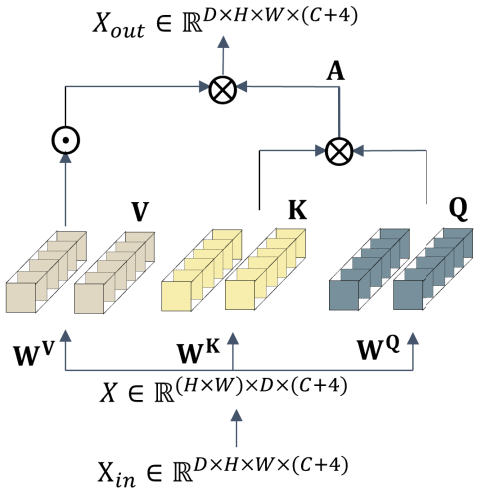


Fig. 8. Structure of the wavelength attention module, enhancing spectral representation by computing attention among target-wavelength features.

4) Wavelength Attention Module: The wavelength attention module (see Fig. 8) focuses on computing attention exclusively among target-wavelength features to enhance spectral representation (see Algorithm 5) [58]. Starting from the input $X_{\text{in}} \in \mathbb{R}^{D\times H\times W\times(C+4)}$, the module reshapes the input to $X \in \mathbb{R}^{(H\times W)\times D\times(C+4)}$. Query (Q), key (K), and value (V) are computed as

$$Q = X \cdot W^Q, \quad K = X \cdot W^K, \quad V = X \cdot W^V \tag{14}$$

where W^Q, W^K, and W^V are learnable weights. Attention weights are calculated as

$$\text{Attention} = \text{Softmax}\left(\frac{QK^{\text{T}}}{\sqrt{d_k}}\right). \tag{15}$$

Finally, the output is

$$X_{\text{out}} = \text{Attention} \cdot V. \tag{16}$$

This output is reshaped back to $X_{\text{out}} \in \mathbb{R}^{D\times H\times W\times(C+4)}$, ensuring that the refined wavelength features retain the original structure. This module refines target-wavelength features by leveraging dependencies among them, improving spectral reconstruction.

5) Feedforward Network: The feedforward network (FFN) (see Fig. 9) serves as a bridge between the encoder and the decoder, transforming the spatial features extracted by the PLM into representations suitable for skip connections. These skip connections directly predict the feature maps of the current target wavelengths, ensuring that essential spectral–spatial details from the encoder are preserved. This design mitigates the information loss during encoding and provides the decoder

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

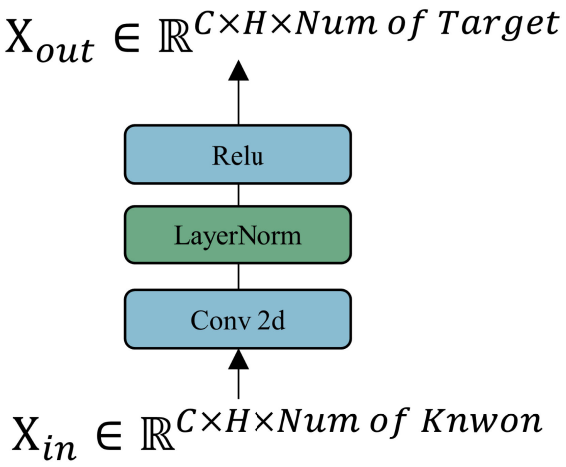


Fig. 9. Structure of the FFN, acting as a bridge between the encoder and the decoder, facilitating feature transformation and skip connections.

with a direct reference to the original spatial features, enhancing the reconstruction of target wavelengths. By leveraging this structure, the FFN facilitates a streamlined flow of relevant information, similar to ResNet's residual connections [59] while focusing on maintaining the integrity of features critical to hyperspectral reconstruction.

6) Integration of Siren for Reconstruction: Siren (sinusoidal representation networks) [60] is integrated into our model as the final reconstruction module due to its superior ability to perform implicit neural mappings. Siren excels at modeling complex relationships between input features and output values, making it particularly suited for tasks involving continuous signals. In our model, each target wavelength's feature vector contains not only its original wavelength value but also enriched contextual information, including its spectral neighborhood and spatial characteristics. This diverse and contextually rich input aligns naturally with Siren's strengths.

By leveraging sinusoidal activation functions, Siren effectively captures high-frequency signal components, which are critical for accurately reconstructing spectral data that exhibit fine-grained wavelength dependencies. Additionally, Siren's capacity to learn implicit mappings ensures a smooth and precise transition from the feature space to the reconstructed spectral values. This makes it an intuitive choice for modeling the nonlinear and intricate relationships required for hyperspectral reconstruction.

The integration of Siren allows our model to fully utilize the spatial and spectral context extracted by earlier modules, enabling a direct and efficient mapping to high-quality reconstructed spectral outputs. This design ensures that the reconstruction process maintains fidelity and precision, aligning with the overall goals of the model.

E. Loss Function

Overall, the model's optimization objective consists primarily of two loss functions: the hyperspectral reconstruction loss $\mathcal{L}_{\mathrm{HS}}$, which measures the error between the model's predicted hyperspectral data and the true hyperspectral data to ensure accurate spectral reproduction, and the MS consistency loss $\mathcal{L}_{\mathrm{MS}}$, which constrains the model to preserve the information from the original MS input during feature extraction, preventing information loss in the U-Net structure. The above formula is as follows:

$$\mathcal{L}_{\mathrm{HS}} = \mathcal{L}_{\mathrm{encoder}} + \mathcal{L}_{\mathrm{decoder}} \tag{17}$$

$$\mathcal{L}_{\mathrm{MS}} = \mathcal{L}_{\mathrm{final\ encoder}}. \tag{18}$$

Within this overall framework, the model further refines the constraints by designing a multilevel loss function. Specifically, in the first $L-1$ encoder blocks, intermediate outputs are directly supervised by comparing the predicted target wavelengths with the corresponding ground truth channels; this early stage supervision, computed using the $L1$ loss, is denoted as

$$\mathcal{L}_{\mathrm{encoder}} = \alpha \sum_{i=1}^{L-1} \mathcal{L}_1\left(O^{(i)}_{\mathrm{encoder}}, G^{(i)}\right). \tag{19}$$

Next, the final output of the encoder is passed through a convolutional operation to regress it back to the dimensions of the original MS input, thereby providing an additional supervisory signal that ensures critical information is retained; this loss is represented as

$$\mathcal{L}_{\mathrm{final\ encoder}} = \beta * \mathcal{L}_1\left(g\left(O_{\mathrm{encoder,final}}\right), G_{\mathrm{input}}\right). \tag{20}$$

Finally, the decoder's output, which represents the fully reconstructed hyperspectral image, is compared with the true hyperspectral data to ensure that the complete spectral distribution is captured, and this loss is expressed as

$$\mathcal{L}_{\mathrm{decoder}} = \gamma * \mathcal{L}_1(O_{\mathrm{decoder}}, G_{\mathrm{full}}). \tag{21}$$

Combining these components, the overall multilevel loss function can be expressed as

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{encoder}} + \mathcal{L}_{\mathrm{finalencoder}} + \mathcal{L}_{\mathrm{decoder}} \tag{22}$$

where $O^{(i)}_{\mathrm{encoder}}$ represents the intermediate outputs of the ith encoder block, $g(O_{\mathrm{encoder,final}})$ is the regressed output from the encoder's final representation, and O_{decoder} is the decoder's output. The corresponding ground truths $G^{(i)}$, G_{input}, and G_{full} are the target channels, original input, and full hyperspectral data, respectively. The loss function $\mathcal{L}_1$ denotes the L_1 loss [61], which computes the mean absolute error (MAE) between predicted values and ground truth, ensuring robust supervision at each level. The coefficients α, β, and γ balance the contributions of the intermediate, MS consistency, and final reconstruction losses.

In the overall optimization objective, the model integrates the hyperspectral reconstruction loss and the MS consistency loss as follows:

$$\arg\min_{\theta}(\mathcal{L}_{\mathrm{total}}). \tag{23}$$

This design ensures that the model not only accurately reconstructs the hyperspectral data but also maintains the integrity of the original MS information through detailed multilevel supervision, thereby enhancing the overall robustness and accuracy of the reconstruction process.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

TABLE I
DISTRIBUTION OF HYPERSPECTRAL IMAGE PATCHES ACROSS TRAINING, VALIDATION, AND TEST SETS

Area	(a)	(b)	(c)	(d)	(e)	(f)	(g)
Patchs	1260	1512	1500	2900	2250	4216	5845

IV. EXPERIMENT

A. Datasets and Experimental Setup

1) Dataset Division: The hyperspectral dataset was divided into training, validation, and test sets. Due to the scarcity of remote sensing data, small sample sizes were used for training, as obtaining hyperspectral data labels is difficult in many areas. This decision reflects the fact that models requiring large sample sizes for training are not ideal for remote sensing scenarios, which prioritize generalization and fine-tuning. Specifically, 10% of the image patches from region (a) were randomly selected for training, and the remaining 90% were used for validation. Regions (b)–(g) were reserved for testing to evaluate the model's ability to generalize. All hyperspectral images were cropped into 16 × 16 pixel patches without spatial overlap. Table I shows the distribution of patches across the subsets.

2) Experimental Detail: All experiments were conducted on a computing system equipped with an Intel i7-13700F CPU and an NVIDIA RTX 4080 GPU. The model was implemented using PyTorch (version 2.1.1) with CUDA 12.1 for GPU acceleration.

To evaluate the model's performance, three widely recognized metrics were used. Mean peak signal-to-noise ratio (MPSNR) quantifies the overall signal-to-noise ratio of the reconstructed hyperspectral images, reflecting the fidelity of the reconstruction [62]. Mean structural similarity index (MSSIM) [63] assesses the structural similarity between reconstructed and reference images, emphasizing spatial accuracy and fidelity. Spectral angle mapper (SAM) [64] evaluates the angular difference between spectral vectors of reconstructed and reference images, capturing the consistency of spectral information. These metrics collectively provide a comprehensive evaluation of the model's performance across spatial, structural, and spectral dimensions.

B. Comparison With Other Methods

To evaluate the performance of our proposed method, we compare it with several state-of-the-art baseline models, including AWAN [39], HDNet [65], HRNet [57], and MST++ [66]. Each baseline represents a unique strategy for hyperspectral image reconstruction. Specifically, AWAN employs an attention mechanism to enhance feature extraction through multilevel fusion. HDNet leverages deep convolutional networks to model the joint spatial and spectral dependencies. HRNet utilizes a multiresolution feature representation approach, emphasizing the importance of high-resolution features during reconstruction. MST++ adopts a multiscale transformer architecture, excelling in capturing long-range dependencies to optimize spectral reconstruction.

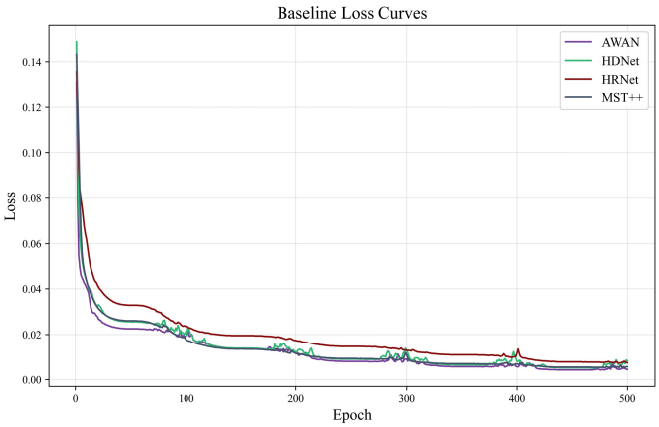


Fig. 10. Training details of all baseline models, ensuring fair and consistent evaluation.

As shown in Fig. 10, all baseline models were trained using the same training set described in Section IV-A, ensuring a fair comparison. The training process followed the recommended settings from their respective original papers, and hyperparameters were adjusted to achieve optimal performance on the validation set. This ensures that all baselines were sufficiently trained and their performance reflects their maximum potential within the scope of our study.

The quantitative results are presented in Tables II–IV, summarizing the MPSNR, MSSIM, and SAM metrics across all test regions. From Table II, our method achieves the highest MPSNR in most regions, particularly in regions (a), (b), (d), and (g), with values of 41.1489, 39.0541, 40.4052, and 39.6420, respectively. These results highlight our method's capability to reconstruct high signal-to-noise ratio images. However, in region (f), AWAN outperforms all other models with the highest MPSNR of 39.8808. Interestingly, despite AWAN's numerical advantage in region (f), its reconstructed images exhibit noticeable color distortions, suggesting that higher MPSNR does not necessarily correlate with better visual quality.

The MSSIM and SAM results, presented in Tables III and IV, respectively, further validate the superior performance of our approach in preserving both visual and spectral consistency. Specifically, as demonstrated in Table III, our method achieves the highest MSSIM values across all test regions, outperforming other baseline models. Moreover, the reconstructed images generated by our approach exhibit negligible color and brightness differences compared to the ground truth across all regions, underscoring the exceptional visual reconstruction capability of our model.

In terms of SAM, the results presented in Table IV further underscore the superior spectral consistency achieved by our method. Our approach consistently produces lower SAM values compared to baseline models, demonstrating its enhanced capability to accurately reconstruct spectral information. This superior performance is critical for spectral-based species analysis, where even minor discrepancies in spectral

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTITATIVE COMPARISON OF MPSNR BETWEEN THE PROPOSED METHOD AND BASELINE MODELS ACROSS TEST REGIONS

Method	(a)	(b)	(c)	(d)	(e)	(f)	(g)
AWAN	38.0447	37.6612	37.8410	39.6957	39.3068	**39.8808**	38.5994
HDNet	39.5805	38.4083	37.0783	39.4325	39.2746	37.3535	38.6246
HRNet	38.6401	37.8219	36.9967	39.3155	39.0863	37.3786	38.1663
MST++	39.8688	39.0400	**38.3549**	40.1060	**39.9639**	38.9738	39.4150
HPWF(ours)	**41.1489**	**39.0541**	38.3514	**40.4052**	38.6576	37.7785	**39.6420**

TABLE III
QUANTITATIVE COMPARISON OF MSSIM BETWEEN THE PROPOSED METHOD AND BASELINE MODELS ACROSS TEST REGIONS

Method	(a)	(b)	(c)	(d)	(e)	(f)	(g)
AWAN	0.9752	0.9770	0.9738	0.9799	0.9799	0.9777	0.9786
HDNet	0.9792	0.9722	0.9683	0.9727	0.9732	0.9765	0.9727
HRNet	0.9748	0.9751	0.9691	0.9753	0.9763	0.9736	0.9743
MST++	0.9784	0.9770	0.9734	0.9786	0.9789	0.9784	0.9777
HPWF(ours)	**0.9880**	**0.9846**	**0.9853**	**0.9809**	**0.9820**	**0.9839**	**0.9820**

TABLE IV
QUANTITATIVE COMPARISON OF SAM BETWEEN THE PROPOSED METHOD AND BASELINE MODELS ACROSS TEST REGIONS

Method	(a)	(b)	(c)	(d)	(e)	(f)	(g)
AWAN	0.1016	0.1198	0.0844	0.0754	0.0862	0.0726	0.0788
HDNet	0.0875	0.1129	0.0923	0.0866	0.0958	0.0780	0.0901
HRNet	0.0866	0.1045	0.0836	0.0774	0.0867	0.0720	0.0805
MST++	0.1162	0.1465	0.0950	0.0819	0.0954	0.0807	0.0884
HPWF(ours)	**0.0575**	**0.0742**	**0.0694**	**0.0632**	**0.0690**	**0.0566**	**0.0644**

angular measurements can significantly impact the results. While baseline methods may exhibit competitive performance in certain areas, they generally incur higher spectral angular errors, likely due to less refined feature extraction strategies.

The visual discrepancies described above are clearly demonstrated in Fig. 11. All baseline models exhibit noticeable distortions in regions where they underperform, whether in overall coloration, contrast, or soil color representation. In particular, in region (f), the baseline models fail to accurately reproduce the soil colors present in the ground truth, as their outputs are significantly influenced by extensive vegetation areas, leading to unrealistic color rendition in nonvegetated regions.

Overall, our method's design, which focuses on target wavelengths at each stage, provides a more structured approach to hyperspectral reconstruction. This targeted modeling likely explains its superior performance in SAM and MSSIM, as it enables the model to capture the intricate relationships among spectral bands while preserving spatial details. Compared to the baseline methods, our approach achieves a more balanced performance, ensuring robust reconstruction across diverse metrics and test regions.

Fig. 12 presents the mean spectral curves across test regions (b)–(g), reconstructed by AWAN, HDNet, HRNet, MST++, and our proposed HPWF-Net, with the ground truth curve as a reference. The x-axis represents the wavelength range (490–840 nm), and the y-axis denotes reflectance values.

In the 700–750-nm range, all models demonstrate a close fit to the ground truth across test regions. This result is largely driven by the concentration of vegetation red-edge features within this spectral interval. Moreover, since forested areas dominate our hyperspectral dataset, their spectral characteristics heavily influence the computed average spectral curves. This predominant representation enhances the availability of training information in this wavelength range, thereby enabling a more effective learning of the mapping between MS and hyperspectral data.

From the perspective of model comparison, both HDNet and HRNet exhibit certain deficiencies in fitting performance compared to other models. Although they successfully capture

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

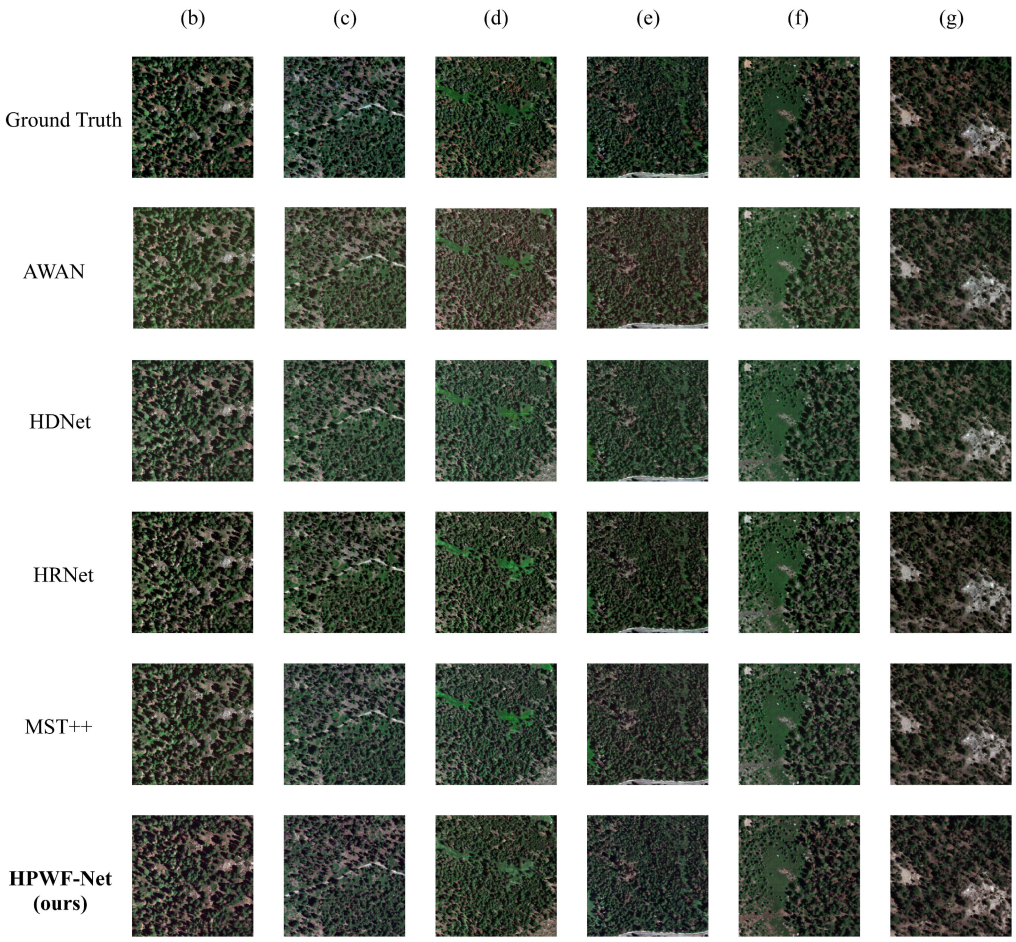


Fig. 11. Qualitative comparison of reconstruction results among baseline models and the proposed method.

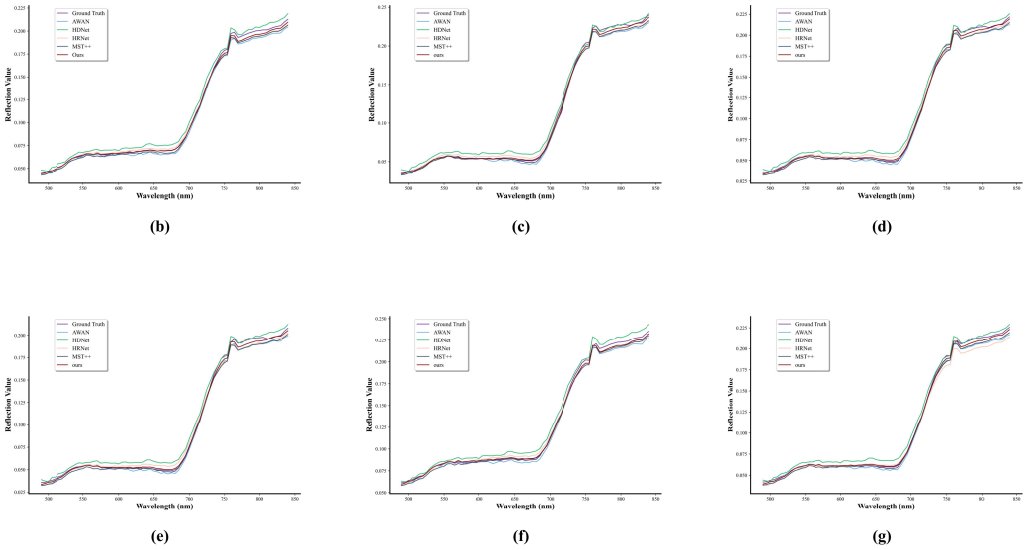


Fig. 12. Mean spectral curves of test regions (b)–(g) reconstructed by baseline models and the proposed HPWF-Net, with ground truth as reference.

the overall trend, they struggle to accurately represent some high-frequency details.

Although MST++ and AWAN demonstrate strong fitting performance over most of the spectral range, their backbone

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

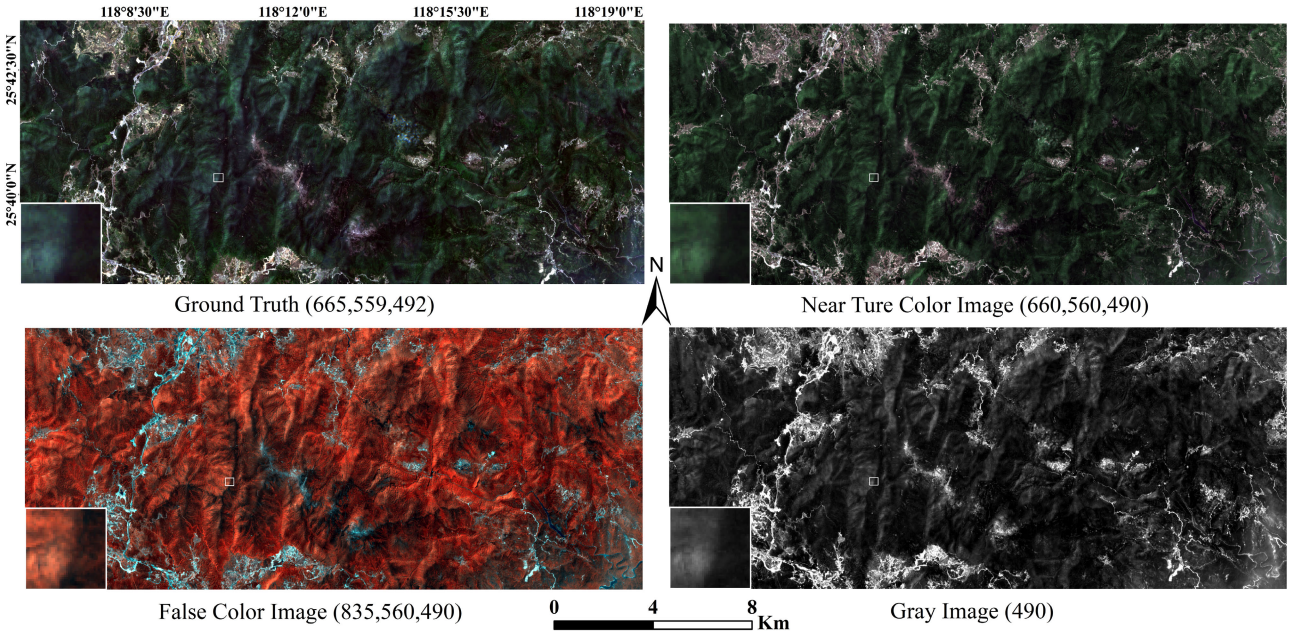


Fig. 13. Reconstructed hyperspectral imagery in near-true color, false color, and grayscale, showing spatial and spectral consistency in the study area. The number in parentheses indicates the image RGB channel wavelength in nanometers.

networks are fundamentally based on convolutional neural networks (CNNs), which inherently prioritize the recovery of image details over the nuanced spectral differences. As a result, their performance in fitting specific spectral bands remains somewhat suboptimal.

In contrast, our proposed HPWF allocates substantial effort in the encoding stage to extract per-pixel spectral features. By integrating an MLP-based Siren architecture for the final reconstruction, HPWF-Net is particularly adept at capturing high-frequency signals. This approach enables the model to preserve subtle spectral variations and intricate details, leading to a more accurate representation of both global trends and local nuances. Consequently, our model demonstrates superior spectral fitting performance compared to the baseline methods.

In summary, from a spectral perspective, HPWF-Net achieves enhanced reconstruction by effectively integrating global trend recovery with the accurate preservation of high-frequency details, thereby overcoming the limitations of the baseline models.

C. Usage in Study Area

To evaluate the practicality and applicability of our proposed method, we applied the pretrained model, originally optimized on the NEON dataset, to the spectral reconstruction of Sentinel-2 imagery within our study area.

Fig. 13 shows the reconstructed hyperspectral imagery using multiple band combinations to assess spectral reconstruction performance. The top-left panel (ground truth) is composed of bands 665, 559, and 492, serving as the reference for subsequent comparisons. The top-right panel presents a near-true color image (660, 560, and 490), closely approximating natural color channels, while the bottom-left panel adopts a false color scheme (835, 560, and 490), highlighting vegetated areas in red. Finally, the bottom-right panel is a single-channel grayscale image (band 490), accentuating structural contrasts such as roads and nonvegetated regions.

Across these representations, the reconstructed forested regions exhibit high spatial and spectral fidelity, as evidenced by the clear delineation of tree canopies and consistent textural details. The near-true color and grayscale images, in particular, underscore the model's ability to capture fine-scale variations in vegetation density and terrain features. Meanwhile, the false color composite vividly accentuates the forested zones in red, further confirming the accurate retrieval of vegetation-related reflectance properties.

These results demonstrate that our reconstruction approach preserves both structural and spectral characteristics in forested areas, suggesting its applicability to high-precision forest monitoring tasks. Nevertheless, discrepancies outside the forest zones remain noticeable. As the pretrained model was developed predominantly using forest-related data, it lacks the specialized training necessary for accurately reconstructing nonforested surfaces. This limitation points to the need for a more diverse training dataset when extending the method to heterogeneous landscapes.

In this study, we rigorously validated the effectiveness of our reconstructed hyperspectral data through a pixel-wise tree species classification task conducted within selected pure forest plots identified by national forest resource inventory data. Because object-based vegetation classification has reached maturity in recent years [67], [68], we deliberately employed a pixel-wise approach to impose stricter demands on spectral fidelity, as even minor reconstruction artifacts become evident at this finer resolution. By representing spectral signatures as structured tabular data, we leveraged CatBoost's proven ability to manage high-dimensional feature interactions. Our dataset (comprising 375 806 georeferenced sample points) was partitioned into training, validation, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

TABLE V
PIXEL-WISE TREE SPECIES CLASSIFICATION RESULTS COMPARING RECONSTRUCTED HYPERSPECTRAL DATA AND ORIGINAL SENTINEL-2 IMAGERY

Dataset	Accuracy	Precision	Recall	F1-Score
Sentinel-2 Imagery	0.5875	0.60	0.54	0.56
Reconstructed Hyperspectral	**0.6465**	**0.65**	**0.62**	**0.63**

testing subsets in an 8:1:1 ratio. Table V compares the classification results using the reconstructed hyperspectral data with those obtained directly from the original Sentinel-2 imagery.

The results demonstrate that the reconstructed hyperspectral data significantly improve classification performance compared to the original Sentinel-2 imagery. Specifically, the overall accuracy increased by approximately 5.9%, with corresponding gains observed in precision, recall, and $F1$-score across all tree species. Notably, the improvements are attributed to the richer spectral information provided by the reconstructed hyperspectral bands, which enhances the model's ability to distinguish subtle spectral variations between tree species.

In summary, the application of our method to the Sentinel-2 imagery within the study area not only achieves high-fidelity spectral reconstruction but also enables more accurate downstream analysis, such as pixel-wise tree species classification. This highlights the practical value of our approach in forest monitoring and ecological studies, particularly in regions with limited access to high-resolution hyperspectral data.

V. CONCLUSION

In this study, we proposed the HPWF-Net, a novel framework for hyperspectral image reconstruction. By progressively focusing on target wavelengths and spatial details, HPWF-Net effectively enhances spectral–spatial consistency. Experimental evaluations demonstrate that HPWF-Net achieves improved results in terms of MPSNR, MSSIM, and SAM metrics. Visual outcomes further illustrate its robustness in forested areas, significantly reducing artifacts and color distortions.

The application of HPWF-Net to Sentinel-2 imagery validated practical utility by substantially improving the accuracy of pixel-wise tree species classification compared to the original MS data. These findings confirm the method's effectiveness in supporting high-precision forest mapping and ecological monitoring. Future research will aim to extend HPWF-Net's applications to more complex ecosystems and other remote sensing scenarios.

REFERENCES

[1] M. P. Krishna and M. Mohan, "Litter decomposition in forest ecosystems: A review," *Energy, Ecol. Environ.*, vol. 2, no. 4, pp. 236–249, Jul. 2017, doi: 10.1007/s40974-017-0064-9.

[2] P. Baldrian, "Forest microbiome: Diversity, complexity and dynamics," *FEMS Microbiol. Rev.*, vol. 41, no. 2, pp. 109–130, Nov. 2016, doi: 10.1093/femsre/fuw040.

[3] R. K. Dixon, A. M. Solomon, S. Brown, R. A. Houghton, M. C. Trexier, and J. Wisniewski, "Carbon pools and flux of global forest ecosystems," *Science*, vol. 263, no. 5144, pp. 185–190, Jan. 1994.

[4] J. Chen et al., "Microclimate in forest ecosystem and landscape ecology," *BioScience*, vol. 49, no. 4, pp. 288–297, Apr. 1999, doi: 10.2307/1313612.

[5] X. Morin, L. Fahse, H. Jactel, M. Scherer-Lorenzen, R. García-Valdés, and H. Bugmann, "Long-term response of forest productivity to climate change is mostly driven by change in tree species composition," *Sci. Rep.*, vol. 8, no. 1, p. 5627, Apr. 2018, doi: 10.1038/s41598-018-23763-y.

[6] F. E. Fassnacht et al., "Review of studies on tree species classification from remotely sensed data," *Remote Sens. Environ.*, vol. 186, pp. 64–87, Dec. 2016, doi: 10.1016/j.rse.2016.08.013.

[7] J. C. White, N. C. Coops, M. A. Wulder, M. Vastaranta, T. Hilker, and P. Tompalski, "Remote sensing technologies for enhancing forest inventories: A review," *Can. J. Remote Sens.*, vol. 42, no. 5, pp. 619–641, Sep. 2016, doi: 10.1080/07038992.2016.1207484.

[8] R. J. Keenan, "Climate change impacts and adaptation in forest management: A review," *Ann. Forest Sci.*, vol. 72, no. 2, pp. 145–167, Jan. 2015, doi: 10.1007/s13595-014-0446-5.

[9] L. Zhang, P. Sun, F. Huettmann, and S. Liu, "Where should China practice forestry in a warming world?" *Global Change Biol.*, vol. 28, no. 7, pp. 2461–2475, Apr. 2022, doi: 10.1111/gcb.16065.

[10] J. C. Doelman et al., "Afforestation for climate change mitigation: Potentials, risks and trade-offs," *Global Change Biol.*, vol. 26, no. 3, pp. 1576–1591, Mar. 2020, doi: 10.1111/gcb.14887.

[11] H. Qin, W. Zhou, Y. Yao, and W. Wang, "Individual tree segmentation and tree species classification in subtropical broadleaf forests using UAV-based LiDAR, hyperspectral, and ultrahigh-resolution RGB data," *Remote Sens. Environ.*, vol. 280, Oct. 2022, Art. no. 113143, doi: 10.1016/j.rse.2022.113143.

[12] X. Liu et al., "Neural network guided interpolation for mapping canopy height of China's forests by integrating GEDI and ICESat-2 data," *Remote Sens. Environ.*, vol. 269, Feb. 2022, Art. no. 112844, doi: 10.1016/j.rse.2021.112844.

[13] L. Liu, N. C. Coops, N. W. Aven, and Y. Pang, "Mapping urban tree species using integrated airborne hyperspectral and LiDAR remote sensing data," *Remote Sens. Environ.*, vol. 200, pp. 170–182, Oct. 2017, doi: 10.1016/j.rse.2017.08.010.

[14] J. Townshend, C. Justice, W. Li, C. Gurney, and J. McManus, "Global land cover classification by remote sensing: Present capabilities and future possibilities," *Remote Sens. Environ.*, vol. 35, nos. 2–3, pp. 243–255, Feb. 1991, doi: 10.1016/0034-4257(91)90016-y.

[15] L. Zeng, B. D. Wardlow, D. Xiang, S. Hu, and D. Li, "A review of vegetation phenological metrics extraction using time-series, multispectral satellite data," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111511, doi: 10.1016/j.rse.2019.111511.

[16] J. Chen et al., "Global land cover mapping at 30 m resolution: A POK-based operational approach," *ISPRS J. Photogramm. Remote Sens.*, vol. 103, pp. 7–27, May 2015, doi: 10.1016/j.isprsjprs.2014.09.002.

[17] N. Yokoya, C. Grohnfeldt, and J. Chanussot, "Hyperspectral and multispectral data fusion: A comparative review of the recent literature," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 5, no. 2, pp. 29–56, Jun. 2017, doi: 10.1109/MGRS.2016.2637824.

[18] G. Vivone, "Multispectral and hyperspectral image fusion in remote sensing: A survey," *Inf. Fusion*, vol. 89, pp. 405–417, Jan. 2023, doi: 10.1016/j.inffus.2022.08.032.

[19] M. P. Ferreira, M. Zortea, D. C. Zanotta, Y. E. Shimabukuro, and C. R. de Souza Filho, "Mapping tree species in tropical seasonal semi-deciduous forests with hyperspectral and multispectral data," *Remote Sens. Environ.*, vol. 179, pp. 66–78, Jun. 2016, doi: 10.1016/j.rse.2016.03.021.

[20] C. Corbane et al., "Remote sensing for mapping natural habitats and their conservation status – new opportunities and challenges," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 37, pp. 7–16, May 2015, doi: 10.1016/j.jag.2014.11.005.

[21] K. Xu et al., "How spatial resolution affects forest phenology and tree-species classification based on satellite and up-scaled time-series images," *Remote Sens.*, vol. 13, no. 14, p. 2716, Jul. 2021, doi: 10.3390/rs13142716.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

[22] J. Zhang, B. Rivard, A. Sánchez-Azofeifa, and K. Castro-Esau, "Intra- and inter-class spectral variability of tropical tree species at la selva, Costa rica: Implications for species identification using HYDICE imagery," *Remote Sens. Environ.*, vol. 105, no. 2, pp. 129–141, Nov. 2006, doi: 10.1016/j.rse.2006.06.010.

[23] S. Zhang et al., "A mapping approach for eucalyptus plantations canopy and single tree using high-resolution satellite images in Liuzhou, China," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4410413, doi: 10.1109/TGRS.2023.3327128.

[24] R. A. Borsoi et al., "Spectral variability in hyperspectral data unmixing: A comprehensive review," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 223–270, Dec. 2021, doi: 10.1109/MGRS.2021.3071158.

[25] P. Wang, B. Bayram, and E. Sertel, "A comprehensive review on deep learning based remote sensing image super-resolution methods," *Earth-Sci. Rev.*, vol. 232, Sep. 2022, Art. no. 104110, doi: 10.1016/j.earscirev.2022.104110.

[26] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024, doi: 10.1109/TPAMI.2024.3362475.

[27] D. Rocchini et al., "Remotely sensed spectral heterogeneity as a proxy of species diversity: Recent advances and open challenges," *Ecol. Informat.*, vol. 5, no. 5, pp. 318–329, Sep. 2010, doi: 10.1016/j.ecoinf.2010.06.001.

[28] D. Wen et al., "Change detection from very-high-spatial-resolution optical remote sensing images: Methods, applications, and future directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 68–101, Dec. 2021, doi: 10.1109/MGRS.2021.3063465.

[29] D. R. A. D. Almeida et al., "Monitoring restored tropical forest diversity and structure through UAV-borne hyperspectral and LiDAR fusion," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112582, doi: 10.1016/j.rse.2021.112582.

[30] A. K. Schweiger et al., "Plant spectral diversity integrates functional and phylogenetic components of biodiversity and predicts ecosystem function," *Nature Ecol. Evol.*, vol. 2, no. 6, pp. 976–982, May 2018, doi: 10.1038/s41559-018-0551-1.

[31] J. Zhang, R. Su, Q. Fu, W. Ren, F. Heide, and Y. Nie, "A survey on computational spectral reconstruction methods from RGB to hyperspectral imaging," *Sci. Rep.*, vol. 12, no. 1, p. 11905, Jul. 2022, doi: 10.1038/s41598-022-16223-1.

[32] E. Omia et al., "Remote sensing in field crop monitoring: A comprehensive review of sensor systems, data analyses and recent advances," *Remote Sens.*, vol. 15, no. 2, p. 354, Jan. 2023, doi: 10.3390/rs15020354.

[33] K. Fotiadou, G. Tsagkatakis, and P. Tsakalides, "Spectral super resolution of hyperspectral images via coupled dictionary learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2777–2797, May 2019, doi: 10.1109/TGRS.2018.2877124.

[34] H.-F. Yan, Y.-Q. Zhao, J. C. Chan, and S. G. Kong, "Spectral super-resolution based on dictionary optimization learning via spectral library," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5400216, doi: 10.1109/TGRS.2022.3229439.

[35] L. Liu, Z. Zou, and Z. Shi, "Hyperspectral remote sensing image synthesis based on implicit neural spectral mixing models," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500514, doi: 10.1109/TGRS.2022.3232705.

[36] W. Leng, X. Han, J. Deng, H. Zhang, W. Li, and W. Sun, "Spectral Super-resolution by using universal and private jointed spectral library and its applications," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5519911, doi: 10.1109/TGRS.2024.3405528.

[37] X. Han, W. Leng, H. Zhang, W. Wang, Q. Xu, and W. Sun, "Spectral library-based spectral super-resolution under incomplete spectral coverage conditions," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516312, doi: 10.1109/TGRS.2024.3392606.

[38] Y. Cai et al., "MST++: Multi-stage spectral-wise transformer for efficient spectral reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 744–754, doi: 10.1109/CVPRW56347.2022.00090.

[39] J. Li, C. Wu, R. Song, Y. Li, and F. Liu, "Adaptive weighted attention network with camera spectral sensitivity prior for spectral reconstruction from RGB images," in *Proc. CVPRW*, Jul. 2020, pp. 462–463, doi: 10.1109/CVPRW50498.2020.00239.

[40] Z. Shi, C. Chen, Z. Xiong, D. Liu, and F. Wu, "HSCNN+: Advanced CNN-based hyperspectral recovery from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 939–947, doi: 10.1109/CVPRW.2018.00139.

[41] B. Arad, R. Timofte, O. Ben-Shahar, Y.-T. Lin, and G. D. Finlayson, "NTIRE 2020 challenge on spectral reconstruction from an RGB image," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2020, pp. 446–447, doi: 10.1109/CVPRW50498.2020.00231.

[42] S.-J. Su, J.-F. Liu, Z.-S. He, S.-Q. Zheng, W. Hong, and D.-W. Xu, "Ecological species groups and interspecific association of dominant tree species in daiyun mountain national nature reserve," *J. Mountain Sci.*, vol. 12, no. 3, pp. 637–646, Jun. 2015, doi: 10.1007/s11629-013-2935-7.

[43] J. Wei and Z. Fan, "Growing stock volume estimation for daiyun mountain reserve based on multiple linear regression and machine learning," *Sustainability*, vol. 14, no. 19, p. 12187, Sep. 2022, doi: 10.3390/su141912187.

[44] T. U. Kampe, "NEON: The first continental-scale ecological observatory with airborne remote sensing of vegetation canopy biochemistry and structure," *J. Appl. Remote Sens.*, vol. 4, no. 1, Mar. 2010, Art. no. 043510, doi: 10.1117/1.3361375.

[45] T U. Kampe et al., "The NEON 2013 airborne campaign at domain 17 terrestrial and aquatic sites in California," NEON Tech. Memorandum Series,, Tech. Rep. TM-005, 2013, doi: 10.13140/2.1.1513.1529.

[46] K. Bereta, H. Caumont, and U. Daniels, "The Copernicus app lab project: Easy access to Copernicus data," in *Proc. EDBT*, Aug. 2019, pp. 501–511.

[47] C.-G. Li and Y.-Y. Jiang, "Development of mobile GIS system for forest resources second-class inventory," *J. Forestry Res.*, vol. 22, no. 2, pp. 263–268, May 2011, doi: 10.1007/s11676-011-0160-1.

[48] A. F. H. Goetz, "Three decades of hyperspectral remote sensing of the Earth: A personal view," *Remote Sens. Environ.*, vol. 113, pp. S5–S16, Sep. 2009, doi: 10.1016/j.rse.2007.12.014.

[49] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890, doi: 10.1109/CVPR.2017.660.

[50] D. Marnerides, T. Bashford-Rogers, and K. Debattista, "Spectrally consistent UNet for high fidelity image transformations," 2020, *arXiv:2004.10696*.

[51] I. C. Neagoe, D. Faur, C. Vaduva, and M. Datcu, "Band reconstruction using a modified UNet for Sentinel-2 images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 6739–6757, 2023, doi: 10.1109/JSTARS.2023.3276912.

[52] N. Audebert, B. Le Saux, and S. Lefevre, "Deep learning for classification of hyperspectral data: A comparative review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 159–173, Jun. 2019, doi: 10.1109/MGRS.2019.2912563.

[53] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015, *arXiv:1502.03167*.

[54] J. He et al., "Spectral super-resolution meets deep learning: Achievements and challenges," *Inf. Fusion*, vol. 97, Sep. 2023, Art. no. 101812, doi: 10.1016/j.inffus.2023.101812.

[55] Q. Zhang, Z. Cui, and X. Niu, "Image segmentation with pyramid dilated convolution based on ResNet and U-net," in *Proc. ICONIP*, Oct. 2017, pp. 364–372, doi: 10.1007/978-3-319-70096-0_38.

[56] Z. Zhou et al., "UNet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 6, pp. 1856–1867, Dec. 2020, doi: 10.1109/TMI.2019.2959609.

[57] Y. Zhao, L.-M. Po, Q. Yan, W. Liu, and T. Lin, "Hierarchical regression network for spectral reconstruction from RGB images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 422–423, doi: 10.1109/CVPRW50498.2020.00219.

[58] A. Vaswani et al., "Attention is all you need," 2017, *arXiv.1706.03762*.

[59] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.

[60] V. Sitzmann, J. N. P. Martel, A. W. Bergman, D. B. Lindell, and G. Wetzstein, "Implicit neural representations with periodic activation functions," 2020, *arXiv:2006.09661*.

[61] B. J. Fubara, M. Sedky, and D. Dyke, "RGB to spectral reconstruction via learned basis functions and weights," in *Proc. CVPRW*, Jun. 2020, pp. 480–481, doi: 10.1109/CVPRW50498.2020.00248.

[62] A. Tanchenko, "Visual-PSNR measure of image quality," *J. Vis. Commun. Image Represent.*, vol. 25, no. 5, pp. 874–878, Jul. 2014, doi: 10.1016/j.jvcir.2014.01.008.

[63] A. Hore and D. Ziou, "Image quality metrics: PSNR vs. SSIM," in *Proc. 20th Int. Conf. Pattern Recognit.*, Aug. 2010, pp. 2366–2369, doi: 10.1109/ICPR.2010.579.

[64] O. A. De Carvalho and P. R. Meneses, "Spectral correlation mapper (SCM): An improvement on the spectral angle mapper (SAM)," in *Proc. Summaries 9th JPL Airborne Earth Sci. Workshop*, vol. 9. Pasadena, CA, USA: JPL publication, 2000, p. 2.

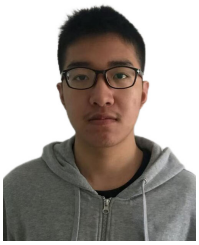
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.

[65] X. Hu et al., "HDNet: High-resolution dual-domain learning for spectral compressive imaging," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 17521–17530, doi: 10.1109/CVPR52688.2022.01702.

[66] Y. Cai et al., "MST++: Multi-stage spectral-wise transformer for efficient spectral reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 745–755, doi: 10.1109/CVPRW56347.2022.00090.

[67] B. Fu et al., "Combination of super-resolution reconstruction and SGA-net for Marsh vegetation mapping using multi-resolution multi-spectral and hyperspectral images," *Int. J. Digit. Earth*, vol. 16, no. 1, pp. 2724–2761, Oct. 2023, doi: 10.1080/17538947.2023.2234340.

[68] B. Fu et al., "Exploring the effects of different combination ratios of multi-source remote sensing images on mangrove communities classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 134, Nov. 2024, Art. no. 104197, doi: 10.1016/j.jag.2024.104197.



Yuhao Yuan received the B.E. degree in new energy materials and devices from China Three Gorges University, Yichang, China, in 2021. He is currently pursuing the M.S. degree with Fujian Agriculture and Forestry University, Fuzhou, China.

His research interests are low-level vision tasks in remote sensing imagery, including super-resolution reconstruction and cloud removal.



Qun Zang received the B.E. degree in computer science and technology from Shandong University of Technology, Zibo, China, in 2024. He is currently pursuing the M.S. degree with Fujian Agriculture and Forestry University, Fuzhou, China.

His research interests include image processing and deep learning.



Zhiping Lin received the B.E. degree in forest engineering from Fujian Agriculture and Forestry University, Fuzhou, China, in 2023, where he is currently pursuing the M.S. degree.

His main research direction is the processing and application of forestry remote sensing data.



Jincheng Liu received the Doctor of Engineering degree in forestry equipment and informatization from Beijing Forestry University, Beijing, China, in 2019.

He is currently an Associate Professor with the Department of Geography Science, Northwest A& F University, Yangling, Xianyang, China. He has published 35 articles in journals, including *ISPRS Journal of Photogrammetry and Remote Sensing*, *Computers and Electronics in Agriculture*, *CATENA*, and *Ecological Indicators*. His research focuses on the development and application of multisource remote sensing and artificial intelligence technologies in areas such as spatiotemporal dynamics of forest vegetation, phenology monitoring, and carbon sequestration assessment.

Dr. Liu serves as an Editorial Board Member/Young Editorial Board Member for international journals, such as *Plant Phenomics*, *Chinese Geographical Science*, *Forestry Research*, and *Agriculture Communications*.



Zhongmou Fan received the Ph.D. degree in forestry equipment and informatization from Beijing Forestry University, Beijing, China, in 2015.

He is currently an Associate Professor with the College of Transportation and Civil Engineering, Fujian Agriculture and Forestry University, Fuzhou, China. His research interests include forestry remote sensing, forest resource investigation and monitoring, and point cloud tree species classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:05:07 UTC from IEEE Xplore. Restrictions apply.